IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: **R26-IT-045**

1. Topic (12 words max)

Machine Learning Based Early Detection of Cattle Diseases

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Cattle diseases are a major problem for the dairy sector as a whole in Sri Lanka due to a lack of facilities for early diagnosis. Serious diseases such as Foot & Mouth Disease (FMD), Lumpy Skin Disease (LSD), Mastitis, and Respiratory infections are usually identified only when clinical signs are apparent and have reached a serious condition. The major stakeholders affected by such a situation are cattle breeders, veterinarians, and any related institutions belonging to the concerned sectors of the Sri Lankan government. The small-scale cattle breeders are mainly affected.

The effects of the delay in disease identification are economic, social, technical, and ethical. Economic implications include loss of regular income from lower dairy production and higher treatment costs. Social implications for dairy-reliant rural areas include the threat of loss of livelihood. From a technical point of view, the main issue with the existing methods of disease identification is the strong manual observation element. From an ethical viewpoint, delay in disease identification leads to unnecessary suffering and poor welfare. In addition, highly contagious diseases like LSD and FMD pose the threat of widespread disease outbreak.

This research is within the purview of CoEAI – Centre of Excellence for Artificial Intelligence under the Artificial Intelligence and Data Science research cluster and falls within the Agriculture and Livestock industry vertical. Machine learning and image-based analysis provide support in early disease detection and decision-making for the proposed solution.

The project addresses the United Nations Sustainable Development Goals, notably SDG 2 by enhancing the resilience of dairy farming; SDG 3 through enhanced practices in animal health management; and SDG 12 by reducing preventable losses to help validate sustainable practices in animal livestock.

References:

- Department of Animal Production and Health, Sri Lanka. (n.d.). *Animal Health Division*. Retrieved from https://daph.gov.lk/divisions/animal_health
- Department of Animal Production and Health, Sri Lanka. (n.d.). *Animal Disease Control Programs*. Retrieved from https://daph.gov.lk/services/an_d
- United Nations. (n.d.). *Sustainable Development Goals*. Retrieved from https://sdgs.un.org

6. Brief description of Existing Research and Systems (300 words max)

The existing solutions related to cattle diseases declaration are basically associated with traditional veterinary practices and farm management. Most farm management solutions support recording the history of health and vaccinations, as well as milk production, but diseases identification relies to a great extent on human observation by farmers and veterinarians, involving actual visits to the farm.

Currently, research has also investigated the use of machine learning (ML) and artificial intelligence (AI) in livestock health. Image processing techniques with convolutional neural networks have also been employed to identify observable signs of livestock diseases like skin lesions, udder inflammation, and hooves. Sensor technologies utilize body temperatures, locomotion, and feeding behavior to detect any abnormalities. Other sophisticated research models use a combination of IoT technologies and ML to produce early warning signals.

Despite the progress made, there are still limitations to existing systems and research. Indeed, many systems require the use of more expensive hardware and continuous sensor streams. Most studies focus only on a single disease or a few symptoms. In fact, there is no encompassing system yet for several diseases that are very common to cows. The early stage of the disease is also less focused on because the models were trained to focus only on the advanced symptoms.

This indicates that there are gaps in the availability of a cost-effective solution that can detect multiple cattle disease conditions using image and health record data. This solution will facilitate the bridging process between the development stage and applications to ensure veterinary intervention and decision-making by farmers and veterinary authorities.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Description of the Solution's Nature:

The proposed system represents a Machine Learning-Based Early Detection System for major cattle diseases, designed to overcome some of the limitations that exist in approaches and research related to the subject. Most of the related works currently deal with single diseases, mostly depend on expensive IoT hardware, and often detect conditions at advanced stages. Our system will integrate four disease-specific modules-Mastitis, Foot, and Mouth Disease (FMD), Lumpy Skin Disease (LSD), and Respiratory Disease providing a scalable, cost-effective, locally adaptable solution suitable for Sri Lankan farms.

Addressing Existing Gaps:

- Multiple Diseases Simultaneous Detection: Each of the modules can target a particular disease.
- Early Detection: Training on subtle, pre-symptomatic indicators of disease using combined visual and numeric data.
- Reduced Dependency on IoT**:** Uses easily accessible data such as smartphone images and manually recorded health parameters.
- Local Adaptability: The model is trained from local region-specific data, and this ensures proper detection according to local farming practices.
- Decision Support: Enabling decision-making with a dashboard and alert system for the farmer and veterinary officers.

System Perspective:

Component 1: Mastitis Detection

- **Input:** Udder images, milk quality metrics, temperature, feed intake
- **Process:**
  - Image preprocessing: resizing, normalization, and noise reduction
  - Extraction of Features Using Convolutional Neural Networks (CNNs)
  - Integration with numeric health parameters
  - Machine learning classification to predict mastitis risk
- **Output:** Risk scoring (low, medium, high), early warning, recommended veterinarian response

Component 2: Foot and Mouth Disease (FMD) Detection

- **Input:** Mouth and hoof images, temperature, activity, and feeding behavior
- **Process:**
  - Image enhancement and region-of-interest detection
  - CNN-based lesion and blister identification
  - Classification model predicts probability of FMD
  - Severity estimation for herd-level risk
- **Output:** Probability percentage, early warning, recommended isolation, and veterinary notification

Component 3: Lumpy Skin Disease (LSD) Detection

- **Input:** Images of the skin (nodules/swellings) Fever Mobility data Behavioral data
- **Process:**
  - Image preprocessing: contrast adjustment, segmentation of skin lesions
  - Feature extraction using deep learning
  - Risk assessment combining visual and numeric indicators
- **Output:** Risk score, early alert, recommendations for veterinary intervention

Component 4: Respiratory Disease Detection

- **Input:** Breathing rate, coughing frequency, temperature, activity rates, and metadata
- **Process:**
  - Data normalization and preprocessing
  - Feature extraction using statistical and ML techniques
  - Classification model predicts respiratory illness probability and severity
- **Output:** Risk score, alert to farmer/vet, monitoring and care instructions

Conceptual Diagram:

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

To effectively develop a Machine Learning-Based Early Detection System for Major Cattle Diseases, there is a requirement of expertise and data that could be acquired through skillful research. To gain necessary expertise in veterinary science, there is interaction with veterinarians and livestock officers, as well as consulting farm managers and relevant official sources such as the Department of Animal Production and Health (DAPH, Sri Lanka). To gain scientific insight into disease pathology and treatment processes, there is a study of relevant scientific literature and research papers.

Technical knowledge is attained by learning programming, machine learning, image processing, and data analysis. The team members learn how to apply Python and machine learning algorithms to develop models for analyzing image and number data related to the health of subjects. Convolutional Neural Networks (CNNs) are applied in detecting the slightest symptoms of a disease in cattle from the taken images, and other classification and feature selection methodologies are applied to the number data related to the body temperature, feeding habits, and respiration rates. The skill in data visualization and dashboard creation is useful in the early warning display for the farmers and the veterinary officers.

Data collection encompasses both primary and secondary sources. In the primary source, the information is gathered by visiting the farm and observing the veterinary centers. In this instance, the information encompasses the images of the cows (udder, skin, mouth, and hooves), as well as the health information that has to be noted down. On the other hand, the secondary source encompasses information obtained from appropriate public platforms. In this context, the information includes the reports from DAPH.

This blend of veterinary knowledge, technical expertise, and high-quality data guarantees that the project team is well-equipped to execute a successful early detection system. The project team strategy will allow them to have a precise, on-time, and feasible detection of a variety of cattle diseases. This will, in turn, have a significant impact on improving cattle management, cattle welfare, as well as cattle productivity in the dairy cattle sector.

9. Objectives and Novelty

Main Objective To develop a **Machine Learning Based Early Detection of Cattle Diseases** that identifies Mastitis, Foot and Mouth Disease (FMD), Lumpy Skin Disease (LSD), and Respiratory Disease at early stages using accessible data such as images and health parameters, providing real-time alerts and decision support for farmers and veterinarians.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Perera L.C.C IT22153036	Mastitis Detection	Collect udder images and health data, preprocess images, extract features using CNN, train classification model, evaluate performance	First implementation of a combined image and numeric-data model for early mastitis detection in Sri Lankan cattle using low-cost data sources
Sudasingha D.D IT22129512	Foot and Mouth Disease (FMD) Detection	Collect mouth and hoof images, gather temperature/activity data, preprocess images, apply CNN for lesion detection, train classifier, validate results	Integrates visual and numeric data for multi-stage FMD detection; early-stage lesion detection not emphasized in prior local research
Manathunga M.A.A.S IT22282422	Lumpy Skin Disease (LSD) Detection	Collect skin images, record health parameters, preprocess and segment lesion areas, extract features, train deep learning model, test and evaluate	Novel localized LSD detection model tailored for Sri Lankan cattle with early-stage visual symptom recognition
Udawaththa M.P.A.M IT22221725	Respiratory Disease Detection	Collect respiratory and activity data, normalize numeric data, extract relevant features, train ML classifier, evaluate risk scoring	Unique integration of physiological and behavioral data for early respiratory disease detection in cattle, addressing small-farm applicability

10. <u>This part is to be filled by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				